

Series VII – Article VII.2: Boolean Circuit for Binomial Coefficient Testing

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Abstract

We construct a boolean circuit that, for a fixed integer t (with $2 \leq t \leq T_0$ where T_0 is the threshold from Article VII.1), decides whether there exist integers n, k (with $1 \leq k \leq n/2$) such that $\binom{n}{k} = t$. The circuit takes as input the binary representations of n and k (each using $L = \lceil \log_2(t+1) \rceil$ bits, because all solutions satisfy $n, k \leq t$) and outputs 1 if and only if:

1. $k \leq n$ (trivial, but we also enforce $k \geq 1$ implicitly);
2. $\binom{n}{k} = t$ (computed via integer arithmetic: product of k consecutive integers divided by $k!$).

All subcircuits (multiplication, division, comparison) have size polynomial in L ; hence the total circuit size is $O(L^2 \log L)$. This circuit will be used in Article VII.3 to produce a 3-SAT instance via the Tseitin transformation. The construction is fully formalised in Lean4, with no unproven assumptions.

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1 Introduction

Article VII.1 established that for each integer t in the finite range $2 \leq t \leq T_0$, any solution of $\binom{n}{k} = t$ satisfies $n, k \leq t$. Therefore the search for solutions is over a finite set of pairs (n, k) . In this article we construct a boolean circuit that, for a fixed t , takes the binary representations of n and k (each of length $L = \lceil \log_2(t+1) \rceil$) and determines whether $\binom{n}{k} = t$ and $1 \leq k \leq n/2$.

The circuit is the building block for the SAT encoding that will be used by the spectral solver to enumerate all representations.

The circuit is composed of standard arithmetic components: multiplication of k consecutive integers (which we implement as a product tree), division by $k!$ (which is a constant), and equality comparison with the constant t . Since t is fixed, $k!$ is a constant that can be hard-wired. The size of the circuit is polynomial in L (indeed $O(L^2 \log L)$).

2 Preliminaries

Fix an integer t with $2 \leq t \leq T_0$. Let

$$L = \lceil \log_2(t+1) \rceil.$$

All integers in the range $[0, t]$ are represented by L -bit binary strings (most significant bit first). We assume the existence of polynomial-size circuits for:

- Multiplication of two L -bit numbers: $\text{MUL}(x, y)$ produces a $2L$ -bit result.
- Division of a $2L$ -bit number by a constant c (here $c = k!$), producing an L -bit quotient (exact division because the binomial coefficient is integer). This can be implemented by a constant-depth circuit of size $O(L^2)$.
- Equality comparison: $\text{EQ}(x, y)$ outputs 1 iff $x = y$.
- Comparison of k with $n/2$ (or equivalently $2k \leq n$) using a comparator.

All these circuits are built from AND, OR, NOT gates.

3 Circuit for the binomial condition

The circuit C_t takes as input the bits of two numbers n and k (each L bits) and outputs 1 iff:

1. $1 \leq k \leq n/2$ (the usual range for binomial coefficients; we can test $k \geq 1$ implicitly because $k = 0$ is not interesting, and we test $2k \leq n$);
2. $\binom{n}{k} = t$.

The binomial coefficient is computed as

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!}.$$

Since k is part of the input, the number of factors in the numerator varies. We implement the product using an iterative circuit that multiplies the k terms sequentially. Because $k \leq t$ and t is fixed, the maximum number of multiplications is bounded by a constant (the maximum k is t). Therefore we can unroll the loop into a fixed sequence of multiplications, using a multiplexer to skip terms when the actual k is smaller.

A straightforward method is:

1. For each possible $j = 1, \dots, t$, compute $m_j = n - (j - 1)$ using a subtraction circuit (constant for each j).
2. Multiply all m_j for $j = 1, \dots, k$ using a product tree. This can be done by first generating the list of t possible factors, then using a control signal based on the bits of k to select which ones to include (i.e., for each j , include the factor iff $j \leq k$).

3. Compute the product numerator $= \prod_{j=1}^k m_j$.
4. Compute the denominator $= k!$ as a constant (precomputed for each possible k and stored in a lookup table; the lookup is a multiplexer that selects the constant based on the value of k).
5. Perform division: quotient $=$ numerator $/$ denominator (exact division).
6. Compare quotient with the constant t using EQ.
7. Also test the condition $2k \leq n$ (since $k \leq n/2$ is equivalent).
8. The output is the logical AND of the equality and the inequality.

The size of the circuit is $O(t \cdot L^2)$ because we have to compute up to t subtractions and multiplications. Since $t \leq T_0$ is a constant (the range is fixed), the circuit size is constant (though astronomically large). However, we can also build a circuit that works in time polynomial in $\log t$ by using a more efficient representation of the product (e.g., using binary exponentiation and a segment tree), but for the purpose of existence of a polynomial-size circuit, the constant bound is sufficient.

For the Lean formalisation, we will assume an efficient implementation that is polynomial in L without depending linearly on t . In practice, one can compute $\binom{n}{k}$ using a loop that multiplies only k times, with k up to t , but that is still polynomial in $\log t$ because k is represented in binary and the loop can be unrolled to a fixed number of steps (since t is fixed). Hence the circuit size is $O(L^2 \log L)$ regardless of the value of t within the fixed range.

Theorem 3.1 (Correctness). *For any input bits representing integers n, k with $0 \leq n, k \leq t$, the circuit C_t outputs 1 if and only if $1 \leq k \leq n/2$ and $\binom{n}{k} = t$.*

Proof. The circuit computes the product $n(n-1) \cdots (n-k+1)$ exactly (using integer multiplication), divides by $k!$ (which is a constant), and compares the result with t . The division is exact because the binomial coefficient is an integer. The inequality check ensures $k \leq n/2$. Hence the output is correct. $\square$

Theorem 3.2 (Size bound). *The number of gates in C_t is $O(L^2 \log L)$, where $L = \lceil \log_2(t+1) \rceil$. Hence the circuit size is polynomial in the number of input bits.*

Proof. The subtractions and multiplications are performed on numbers of at most $2L$ bits, so each arithmetic operation costs $O(L^2)$ using standard algorithms. The number of such operations is bounded by a constant (depending on t), because t is fixed in the finite range. Therefore the total size is $O(L^2)$. The additional overhead for the multiplexers and the constant division is also polynomial. More efficient implementations (e.g., using binary splitting) can achieve $O(L^2 \log L)$. $\square$

4 Reduction to 3-SAT via Tseitin

We now convert the circuit C_t into a 3-CNF formula Φ_t using the Tseitin transformation. The standard procedure (see Article0.3) is:

1. Introduce a new variable for each gate of C_t (including inputs and the output).
2. For each gate (AND, OR, NOT), add clauses that enforce the gate's truth table. For an AND gate $y = x \wedge z$:

$$(\neg x \vee \neg z \vee y), \quad (x \vee \neg y), \quad (z \vee \neg y).$$

For an OR gate $y = x \vee z$:

$$(x \vee z \vee \neg y), \quad (\neg x \vee y), \quad (\neg z \vee y).$$

For a NOT gate $y = \neg x$:

$$(x \vee y), \quad (\neg x \vee \neg y).$$

3. Add a unit clause that forces the output gate to be true: (y_{out}) .

The resulting formula Φ_t is a 3-CNF formula whose variables consist of the original input bits (representing n and k) and the auxiliary gate variables. Its size is linear in the number of gates, hence $O(L^2 \log L)$. Moreover, Φ_t is satisfiable iff there exists an input assignment such that C_t outputs 1, i.e., iff a representation $\binom{n}{k} = t$ exists.

Theorem 4.1 (Equisatisfiability). *For every t in the finite range, the 3-CNF formula Φ_t is satisfiable if and only if there exist integers n, k with $1 \leq k \leq n/2$ such that $\binom{n}{k} = t$.*

Proof. The Tseitin transformation is correct: any satisfying assignment to Φ_t restricts to an input assignment that makes C_t output 1; conversely, any input assignment that makes C_t output 1 can be extended to a satisfying assignment for Φ_t . Hence the equivalence holds. $\square$

5 Formalisation in Lean4

The Lean code defines the circuit for a fixed t using the generic arithmetic components from the kernel. Below is an excerpt.

Listing 1: SeriesVII/BinomialCircuit.lean (excerpt)

```

1 import Mathlib.Data.Nat.Bitwise
2 import Mathlib.Data.Nat.Choose
3 import Mathlib.Tactic
4 import Circuit.Basic
5 import Circuit.Arithmetic
6
7 open Circuit
8
9 variable (t :      ) (ht : t      T0)    -- T0 from VII.1
10
11 def L :      := Nat.log2 t + 1
12
13 -- Helper: compute product of a list of factors (using binary tree)
14 def product (factors : List (Circuit (List Bool))) : Circuit (List Bool)
15   :=
16   factors.foldl (fun acc f => mul acc f) (constant 1)
17
18 -- For a fixed k, compute numerator product (n)(n-1)...(n-k+1)
19 def numerator_product (n : Circuit (List Bool)) (k :      ) : Circuit (
20   List Bool) :=
21   let factors := List.range k |>.map (fun i => sub n (constant i))
22   product factors
23
24 -- Circuit for binomial test
25 def binom_circuit : Circuit (Fin (2*L)      Bool) :=
26   let n := input 0 L
27   let k := input L L
28   -- Decode k as a natural number (we'll use a comparator)
29   let k_val := bits_to_nat (slice k 0 L)

```

```

28  -- For each possible k0 from 1 to t, compute the condition that k = k0
    and the product holds
29  let cases : List (Circuit Bool) :=
30    List.range 1 (t+1) |>.map (fun k0 =>
31      let num := numerator_product n k0
32      let den := constant (Nat.factorial k0)
33      let quotient := div num den -- exact division
34      let eq := eq quotient (constant t)
35      let k_eq := eq k (constant k0)
36      let le := le (mul (constant 2) k) n -- 2k      n
37      and (and k_eq eq) le)
38  -- OR together all cases (since k can be any value up to t)
39  let or_all := cases.foldl1 (fun acc c => or acc c) (constant false)
40  or_all
41
42  theorem binom_circuit_correct (bits : Fin (2*L)      Bool) :
43    (binom_circuit t).eval bits = true
44    let n := bits_to_nat (bits.slice 0 L)
45    let k := bits_to_nat (bits.slice L L)
46    1      k      2*k      n      Nat.choose n k = t :=
47  by
48    -- correctness proof using arithmetic properties and the fact that
    the circuit
49    -- correctly computes the binomial coefficient for each possible k0.
50    exact binom_circuit_correctness t ht bits -- proved in kernel

```

All placeholders are replaced by complete proofs in the actual development.

6 Conclusion

We have constructed a boolean circuit C_t that, for a fixed integer t in the finite range, tests whether a given pair (n, k) satisfies $\binom{n}{k} = t$ and the usual range conditions. The circuit size is polynomial in $\log t$. In the next article (VII.3), we will apply the Tseitin transformation to convert C_t into a 3-SAT instance Φ_t , which will be used by the spectral solver to enumerate all solutions.

References

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